

Soft forward, support-only value gradient

Research note: does withholding the tail’s value gradient recover the hard-forward surrogate?

2026-09-27.

1 Purpose and summary of conclusions

The soft-forward note (`docs/rblapsum-soft-forward.tex`) left a why open: under the hard Top- K evaluation, the soft-forward model loses badly to its hard-forward twin, and one candidate explanation is training-signal dilution – the optimizer trains all $K+J$ feature values at once instead of squeezing the most out of the K it will keep, as the hard-forward surrogate does. This note tests that explanation directly. The regime (`rblapsum_sf_value_grad="support"`, “sf-vsup”) keeps the soft forward $y_i = z_i p_i$ over the pool unchanged, but the J inactive candidates receive no value gradient: only the score-path (support-change) term reaches them, so their ranking still trains while their content does not. Active features keep the soft value gradient $u p_i$; the boundary routing is `through_rank_kappa` with support scale 1.0, as everywhere.

If dilution were the mechanism, removing the tail’s value training should recover a substantial part of the hard-eval gap to the hard-forward twin. The measured answer is that it recovers essentially none of it:

- **The hard-eval gap to the hard-forward twin persists in full.** At the comparison step the gap is +0.045 to +0.909 nats across the grid – the same magnitudes, the same growth with J , and the same inverse ordering in K as the unrestricted soft-forward family. Withholding the tail’s value gradient does not make the hardened model better.
- **The ranking gradient alone still populates the tail.** The trained forward carries 3–82 units of probability mass beyond K despite the tail’s values never receiving task gradient: the score path by itself moves tail features into the kernel’s reach. The hard-eval damage again tracks that mass within each K .
- **Tail value training was a small benefit, not a cost.** Under soft evaluation sf-vsup is slightly *worse* than the unrestricted soft forward in every directly comparable cell (+0.010 to +0.024 nats), while still beating the pool-matched hard Top- K' baseline everywhere (−0.059 to −0.103).
- **Instability is unchanged, placing it in the score path.** (128, 128) at $T = 2$ diverged at step 3520; the unrestricted soft forward died at 3480 in the same cell, and the hard-forward twin survives it. The value-gradient mask is irrelevant to the divergence, consistent with the boundary-mechanism analysis of the kappa-ablation note.
- **One local exception.** At the two largest- K $T = 2$ cells the mask reduces the late hard-eval drift the unrestricted family shows: (256, 256) ends 0.222 and (128, 384) 0.109 nats better than sf at the same step, both still far above kappa. An observation about two cells, not a mechanism.

Taken together: the hard-eval loss of the soft forward is carried by the forward-context mismatch itself – the model is trained with weighted tails present and evaluated with them deleted – and not by where the value gradient goes.

2 Setup

Twenty runs, identical to the soft-forward campaign in every respect but two. The ten (K, J) cells and both variants ($T = 2$; post-norm at $T = 1$) inherit the same hard-forward twins' `config.json` files with the mode fields overridden, on the same machine and repo state (`rb_lapsum_sf_value_grad="support"` added in commit `1dd3764`). First, the value-gradient mask above. Second, the horizon: the runs stop at 10k of the configured 20k steps, with `max_steps` – and with it the learning-rate cosine and every schedule – untouched (`train_guard --stop-step`), so the trajectory up to the stop is step-for-step comparable with the full-length references. All comparisons in this note are therefore read at **step 9500**, the last validation step common to every run: the stop fires on the step-10000 training log, which precedes that step's validation pass in twelve of the twenty runs.

References, both at their own step-9500 points: the hard-forward `through_rank_kappa` twins, and the five hard Top- K' post-norm baselines. Metrics as before: `val/ce` is the hard Top- K forward, `val_soft/ce` the trained soft forward, `rb_soft_mass` the pool's probability mass $\sum_i p_i$.

3 Results

Hard evaluation

`val/ce` at step 9500, sf-vsuf against its hard-forward twin:

(K, J)	$T = 2$			post-norm		
	kappa	sf-vsuf	gap	kappa	sf-vsuf	gap
(32, 32)	1.6758	1.7470	+0.071	1.6591	1.7590	+0.100
(32, 96)	1.6374	1.9294	+0.292	1.6389	1.8749	+0.236
(32, 224)	1.6114	2.1871	+0.576	1.6211	2.0758	+0.455
(32, 480)	1.6218	2.5308	+0.909	1.6717	2.2258	+0.554
(64, 64)	1.6403	1.6967	+0.056	1.6077	1.6839	+0.076
(64, 192)	1.6156	1.8370	+0.221	1.5822	1.7880	+0.206
(64, 448)	1.6006	2.0097	+0.409	1.5773	1.9270	+0.350
(128, 128)	1.6117	diverged (3520)		1.5700	1.6239	+0.054
(128, 384)	1.5874	1.7647	+0.177	1.5556	1.7286	+0.173
(256, 256)	1.5773	1.6809	+0.104	1.5442	1.5891	+0.045

The pattern of the soft-forward note is reproduced unchanged: the loss grows with J at every K , shrinks with K , and the post-norm is less damaged than $T = 2$ at wide windows. Masking the tail's value gradient has not moved it.

Soft evaluation

`val_soft/ce` at step 9500, grouped by pool, against hard Top- K' at the same step:

pool K'	(K, J)	sf-vsuf $T=2$	sf-vsuf post-norm	hard Top- K'	best margin
64	(32, 32)	1.5690	1.6049	1.6718	−0.103
128	(32, 96)	1.5356	1.5530	1.6267	−0.091
	(64, 64)	1.5351	1.5470		−0.092
256	(32, 224)	1.5193	1.5312	1.5928	−0.074
	(64, 192)	1.5099	1.5214		−0.083
	(128, 128)	—	1.5152		−0.078
512	(32, 480)	1.5009	1.5224	1.5597	−0.059
	(64, 448)	1.5027	1.5080		−0.057
	(128, 384)	1.5104	1.5032		−0.056
	(256, 256)	1.5137	1.5095		−0.050

Every cell still beats the pool-matched hard baseline, $T = 2$ still leads the post-norm almost everywhere, widening J still helps monotonically, and K still barely matters at fixed pool – the soft-evaluation regularities of the unrestricted family survive the mask intact.

Against the unrestricted soft forward

At step 9500, sf-vsuf minus sf, same cell and variant (positive = the mask hurt):

(K, J)	hard eval		soft eval	
	$T=2$	post-norm	$T=2$	post-norm
(32, 32)	−0.014	+0.015	+0.011	+0.024
(32, 96)	−0.008	+0.011	+0.018	+0.016
(32, 224)	+0.039	+0.091	+0.023	+0.008
(32, 480)	+0.158	+0.191	+0.015	+0.005
(64, 64)	−0.022	+0.005	+0.010	+0.013
(64, 192)	−0.034	+0.012	+0.009	+0.018
(64, 448)	−0.015	+0.055	+0.023	+0.012
(128, 128)	—	+0.003	—	+0.008
(128, 384)	−0.109	+0.013	−0.051	+0.009
(256, 256)	−0.222	+0.005	+0.016	+0.008

Three readings. Under soft evaluation the mask costs a small, uniform amount (+0.008 to +0.024; the one negative entry is the window of sf’s recovered excursion at (128, 384), $T = 2$, which contaminates the sf reference there) – the tail’s value training was contributing quality, not diluting it. Under hard evaluation the differences are small and two-signed at $J \leq 3K$, and the mask clearly *hurts* at the widest $K = 32$ windows. The two large negative entries ($T = 2$ at (256, 256) and (128, 384)) are the late-drift exception flagged in the summary.

Tail mass without tail value training

rb_soft_mass − K at the end of each run:

(K, J)	$T=2$	post-norm	(K, J)	$T=2$	post-norm
(32, 32)	3.9	2.9	(64, 448)	81.5	29.4
(32, 96)	15.8	8.2	(128, 128)	—	9.0
(32, 224)	35.2	16.3	(128, 384)	75.6	27.4
(32, 480)	72.3	27.9	(256, 256)	23.5	15.3
(64, 64)	7.7	5.2			

The tails carry substantial mass even though their values receive no task gradient: the score path alone – the ranking signal, including the κ -distributed boundary term – moves inactive features into the kernel’s reach, and the hard-eval gap again grows with this mass within each K . This is the mechanical reason the mask cannot close the gap: the forward’s tail contribution, which hard evaluation deletes, is built by the part of the gradient the mask keeps.

Stability

(128, 128) at $T = 2$ diverged at step 3520 (best CE 1.72, then the gradient-norm blowup; guard ceiling rule), against 3480 for the unrestricted soft forward and survival for the hard-forward twin. The value-gradient mask neither delays nor prevents the divergence, so the instability is carried by the score path. All other cells reached the horizon; no recovered excursions this time.

Figures

Layout as in the soft-forward note: per pool, left $T = 2$, right post-norm; hard-forward twins in blues, sf-vsups in oranges (light to dark with K), hard Top- K' dashed black; vertical axis the hard-eval val/ce; the horizon is 10k steps. The soft-eval figure repeats val_soft/ce for the sf-vsups family against the same ladder.

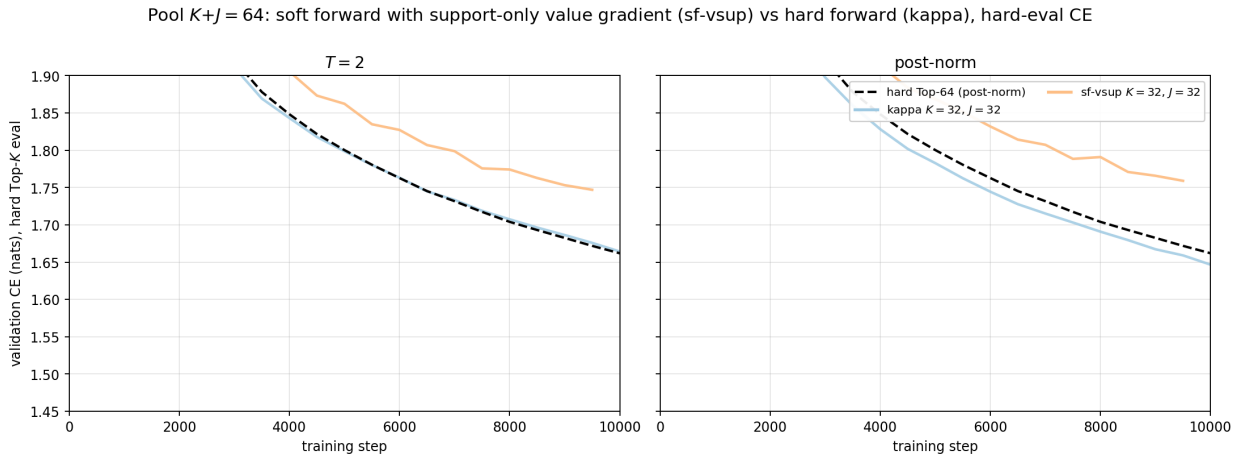


Figure 1: Pool $K' = 64$. The single cell (32, 32) sits 0.07–0.10 nats above its twin under hard eval – the same offset the unrestricted soft forward shows here.

4 Interpretation

The experiment separates two explanations the soft-forward note could not: that the soft forward’s hard-eval loss comes from diluting the training signal across $K+J$ feature values, or from the forward-context mismatch – active features are trained in the presence of weighted tails that hard evaluation then deletes. Removing the tail’s value gradient eliminates the dilution channel by construction, and the hard-eval gap does not move. The mismatch explanation is left carrying the result, and the tail-mass table shows why it must: the ranking signal alone rebuilds the tail contribution, so the trained forward still depends on features the hard forward drops.

The mask’s measurable effects are all second-order and both-signed. It costs a small uniform amount of soft-eval quality (the tail values were learning something useful), it hurts hard eval at small K with very wide windows (with only the ranking signal, the tail that does form is less coordinated with the actives, and at $K = 32$, $J = 480$ the forward leans on it heavily), and it

Pool $K+J = 128$: soft forward with support-only value gradient (sf-vsups) vs hard forward (kappa), hard-eval CE

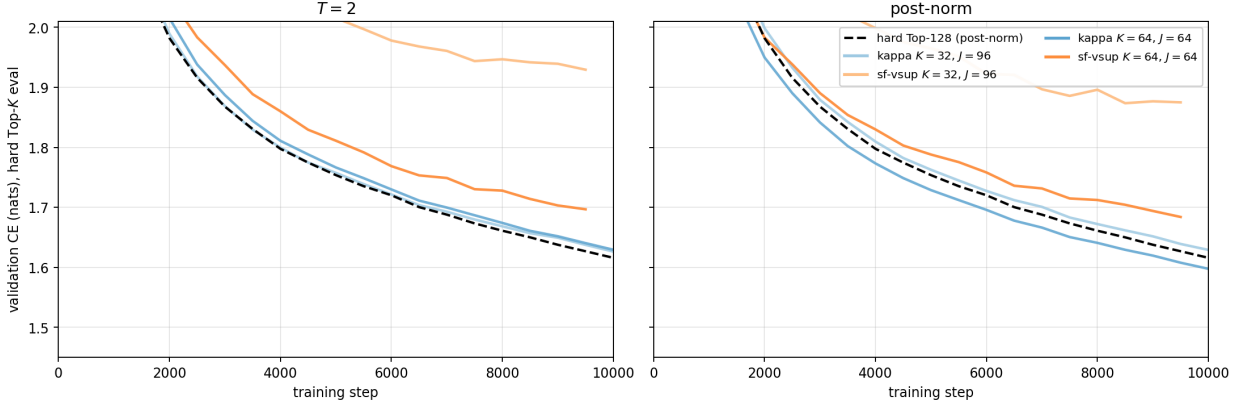


Figure 2: Pool $K' = 128$. Both cells trail their twins by 0.06–0.24 nats; the ordering and the J -dependence match the unrestricted family.

Pool $K+J = 256$: soft forward with support-only value gradient (sf-vsups) vs hard forward (kappa), hard-eval CE

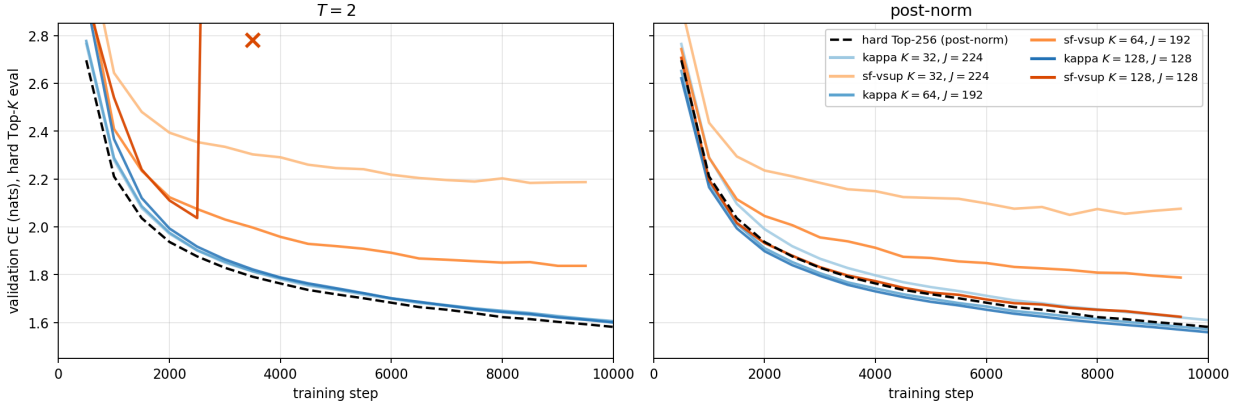


Figure 3: Pool $K' = 256$. The $T = 2$ cell at (128, 128) ends at its guard stop (cross), the second soft-forward death of this cell at 3.5k steps.

damps the late hard-eval drift of the two largest- K $T = 2$ cells. None of this changes any conclusion of the soft-forward note.

For the programme, the negative result is the useful part: closing the train/inference gap is not a matter of where the value gradient goes. The remaining levers are the ones named before – annealing T toward a hard forward, a hardening fine-tune, or a penalty on `rb_soft_mass` – all of which act on the forward’s tail mass itself rather than on the gradient’s routing.

5 Scope

Single runs at one seed, and a 10k-step horizon on the 20k schedule: all numbers are mid-training values at step 9500, directly comparable across families because every run shares the schedule, but not comparable with the 20k finals of the earlier notes. The unrestricted-sf comparison reuses that campaign’s runs at the same step and machine. The hard-forward references remain the retired machine’s runs. The (128, 384), $T = 2$ sf reference at step 9500 sits inside that run’s recovered excursion, so the one cell where the mask appears to help soft eval is an artifact of the reference, not of the mask. Only `through_rank_kappa` at support scale 1.0 was trained, and the mask was tested only at the two extremes (all pool values train, or only the support’s); a fractional version was not run.

Pool $K+j = 512$: soft forward with support-only value gradient (sf-vsups) vs hard forward (kappa), hard-eval CE

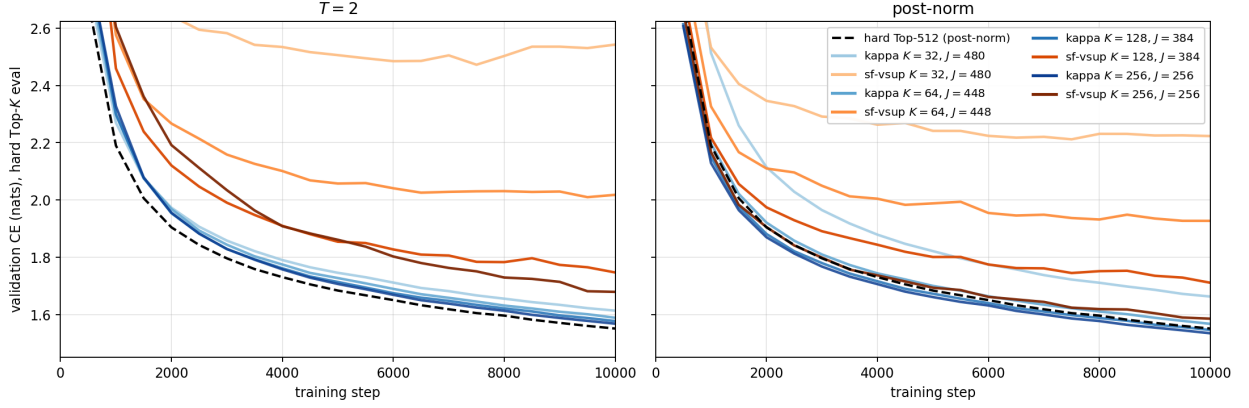


Figure 4: Pool $K' = 512$. The widest windows: the orange family sits 0.05–0.9 nats above the blues under hard eval. Unlike the unrestricted soft forward, the $T = 2$ cells at $K = 128$ and 256 do not drift upward late – the one visible effect of the value-gradient mask.

sf-vsups under its own (soft) forward: val_soft/ce against the hard Top- K' ladder

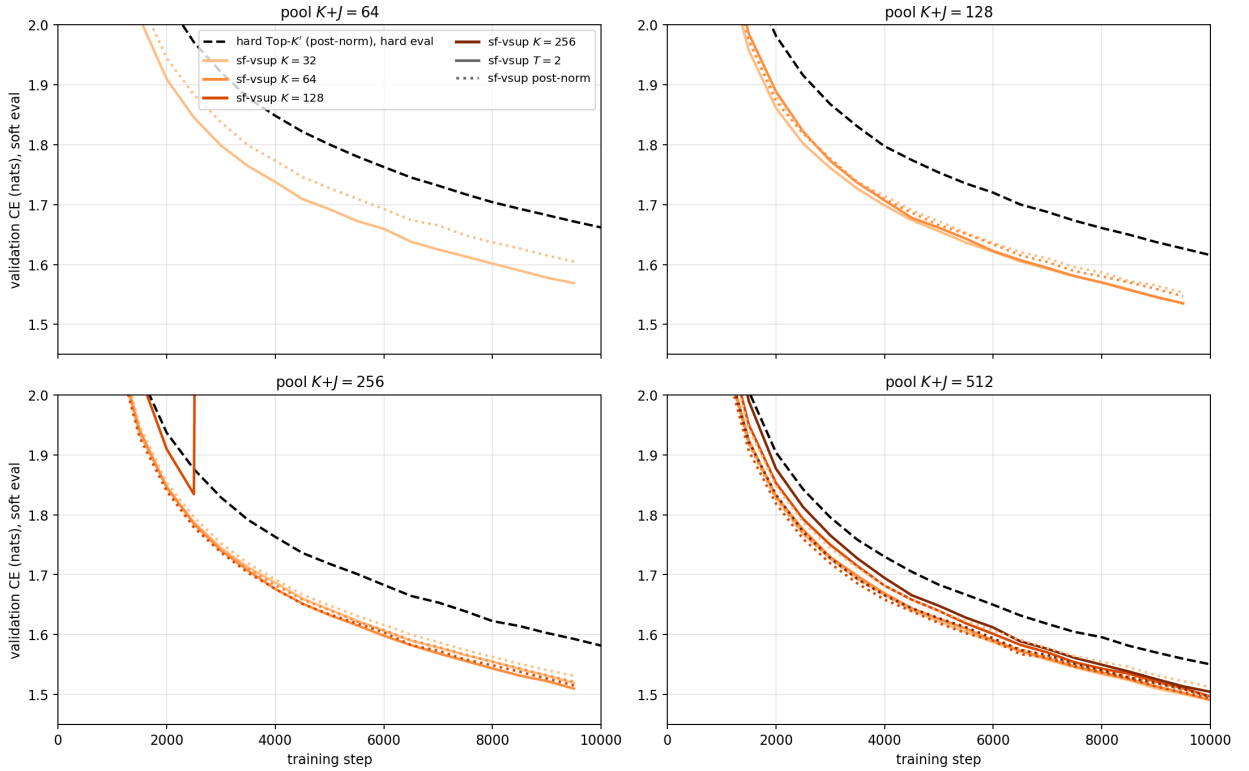


Figure 5: sf-vsups under its own forward (val_soft/ce), per pool, against the pool-matched hard Top- K' baseline (dashed). Solid $T = 2$, dotted post-norm. Every curve ends below the dashed reference; the cluster across K is tight, as in the unrestricted family.